

P&S, Spring 2023, midterm exam

Version 1



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI  
INHA UNIVERSITY IN TASHKENT

FIRST NAME:

LAST NAME:

ID NUMBER:

SECTION:

SIGNATURE:

ROOM NUMBER:

MIDTERM EXAMINATION

COURSE NAME:

PROBABILITY AND STATISTICS

COURSE NUMBER:

MSC3010

EXAMINATION DURATION:

1 HOUR

ADDITIONAL MATERIALS  
ALLOWED TO BE USED:

SIMPLE CALCULATOR

SPECIAL INSTRUCTIONS

(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS.  
(2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET,  
YOU MUST PUT ONLY YOUR **ID** ON EVERY ODD PAGE  
(3) For each question write your solution in details

- Final answers must be written by only blue or black, non-erasable pen.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

| Do NOT fill<br>max points | Q1<br>5 | Q2<br>5 | Q3<br>5 | Q4<br>5 | Q5<br>5 | Q6<br>5 | Total<br>30 |
|---------------------------|---------|---------|---------|---------|---------|---------|-------------|
| your<br>score             |         |         |         |         |         |         |             |

Question 1:

Two cards are drawn from a pack of 52 cards. Is it more likely that two hearts will be drawn when the drawing is with replacement or without replacement? Why?

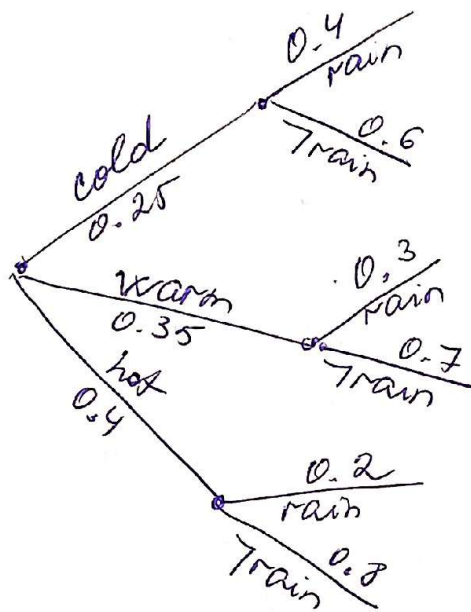
With replacement:  $p = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$

Without replacement:  $p = \frac{1}{4} \cdot \frac{12}{51}$  or  
 $p = \frac{C_2^{13}}{C_2^{52}} = \frac{13!}{2!11!} : \frac{52!}{2!50!} = \frac{13 \cdot 12}{52 \cdot 51} = \frac{3}{51}$

Since  $\frac{1}{16} = \frac{3}{48} > \frac{3}{51}$ , it is more likely to get two hearts with replacement.

Question 2:

The weather on a particular day is classified as either cold, warm, or hot. There is a probability of 0.25 that it is cold and a probability of 0.35 that it is warm. In addition, on each day it may either rain or not rain. On cold days there is a probability of 0.40 that it will rain, on warm days there is a probability of 0.30 that it will rain, and on hot days there is a probability of 0.20 that it will rain. If it is raining on a particular day, what is the probability that it is hot?



$$p(\text{Hot}|\text{rain}) = ?$$

$$p(H|R) = \frac{p(H \cap R)}{p(R)} =$$

$$= \frac{p(R|H) p(H)}{p(R)} =$$

$$= \frac{p(R|H) p(H)}{p(R|H) p(H) + p(R|W) p(W) + p(R|C) p(C)}$$

$$= \frac{0.2 \cdot 0.4}{0.2 \cdot 0.4 + 0.3 \cdot 0.35 + 0.4 \cdot 0.25}$$

$$= \frac{0.08}{0.08 + 0.105 + 0.1} =$$

$$= \frac{0.08}{0.285} \approx \frac{80}{285} \approx 0.28$$



Question 3:

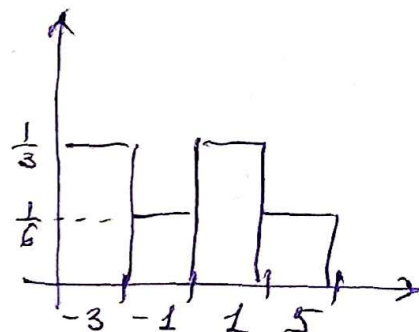
In a game a player rolls a fair six-sided die. If the score is even, the player receives an amount of dollars equal to the score. If the score is odd, the player receives an amount of dollars equal to the score multiplied by two. It costs the player \$5 to play the game.

- (a) What is the probability mass function of the player's net winnings (that is, winnings minus the \$5 payment)?  
(b) Find expected value of the net winning.  
(c) Find variance of the net winning.

even - score  
odd - score \* 2  
cost - 5 \$

Let  $X$  be net winning

|       |               |               |               |               |               |               |
|-------|---------------|---------------|---------------|---------------|---------------|---------------|
| score | 1             | 2             | 3             | 4             | 5             | 6             |
| $X$   | -3            | -3            | 1             | -1            | 5             | 1             |
| prob  | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |



(a)  $\Rightarrow X \in \{-3, -1, 1, 5\}$

$p(X = -3) = p(X = 1) = \frac{1}{3}, p(X = -1) = p(X = 5) = \frac{1}{6}$

(b)  $E(X) = -3 \cdot \frac{1}{3} - 1 \cdot \frac{1}{6} + 1 \cdot \frac{1}{3} + 5 \cdot \frac{1}{6} =$   
 $= -1 - \frac{1}{6} + \frac{1}{3} + \frac{5}{6} = -\frac{7}{6} + \frac{7}{6} = 0$

(c)  $Var(X) = E(X^2) - E(X)^2 =$   
 $= 9 \cdot \frac{1}{3} + \frac{1}{6} + \frac{1}{3} + 25 \cdot \frac{1}{6} - 0 =$   
 $= 3 + \frac{14}{3} = \frac{23}{3}$

Question 4:

Suppose that the random variables  $X$  and  $Y$  have a joint probability density function

$$f(x, y) = x(3 - y)$$

for  $0 \leq x \leq 2$  and  $0 \leq y \leq 1$ .

(a) What is the marginal probability density function of  $X$ ?

(b) Are the random variables  $X$  and  $Y$  independent?

(c) What is  $\text{Cov}(X, Y)$ ?

$$\begin{aligned} (a) \quad f_X(x) &= \int_0^1 x(3-y) dy = x \left[ 3y - \frac{y^2}{2} \right]_0^1 = \\ &= x \left( 3 - \frac{1}{2} \right) = \frac{5}{2}x \end{aligned}$$

(b) For  $X$  and  $Y$  to be independent we must have  $f(x, y) = f_X(x) f_Y(y)$ ,  $\forall x, y$   
So we also need

$$f_Y(y) = \int_0^2 x(3-y) dx = (3-y) \left[ \frac{x^2}{2} \right]_0^2 = 2(3-y)$$

Checking independence:  $f_X(x) f_Y(y) = 5x(3-y) \neq f(x, y)$   
 $\Rightarrow$  not independent.

$$(c) \quad \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$E(X) = \int_0^2 x \cdot f_X(x) dx = \int_0^2 x \cdot \frac{5}{2}x dx = \frac{5}{2} \left[ \frac{x^3}{3} \right]_0^2 = \frac{20}{3}$$

$$E(Y) = \int_0^1 y f_Y(y) dy = \int_0^1 2y(3-y) dy = 2 \left[ \frac{3y^2}{2} - \frac{y^3}{3} \right]_0^1 = \frac{7}{3}$$

$$\begin{aligned} E(XY) &= \int_0^2 \int_0^1 xy f(x, y) dx dy = \int_0^2 \int_0^1 xy(3-y) dx dy = \int_0^2 y(3-y) dy \int_0^1 x^2 dx = \\ &= \left[ \frac{3y^2}{2} - \frac{y^3}{3} \right]_0^1 \left[ \frac{x^3}{3} \right]_0^1 = \frac{7}{6} \cdot \frac{8}{3} = \frac{28}{9} \end{aligned}$$

$$\text{Cov}(X, Y) = \frac{28}{9} - \frac{20}{3} \cdot \frac{7}{3} = \frac{28 - 140}{9} = \frac{-112}{9}$$

Question 5:

In a typical sports playoff series, two teams play a sequence of games until one team, the eventual winner, has won four games. Suppose that in each game team B beats team A with a probability of 0.6, and that the results of different games are independent.

(a) Explain how the negative binomial distribution can be used to analyze this problem.

(b) What is the probability that team B wins the series in game seven?

(a) Winner = win 4 games

We call "B beats A" a success

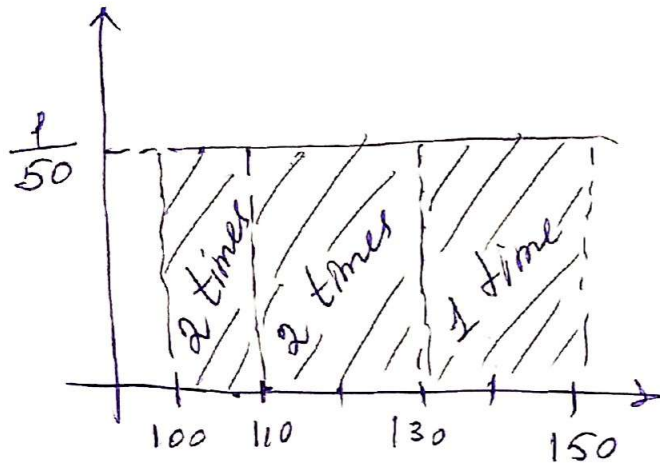
then  $p = 0.6$ , so when we need  $r = 4$

$$\begin{aligned} (b) \quad & p(\text{B wins in game 7} = \text{B wins 4 times in 7th game}) = \\ & = p(\text{B wins exactly 4 games out of 7}) = \\ & = p(X = 7) = \binom{6}{3} \cdot 0.4^3 \cdot 0.6^4 = \\ & = 60 \cdot 0.4^3 \cdot 0.6^4 \approx 0.4977 \end{aligned}$$



Question 6:

Suppose that after operation, the electrical charge remaining on a circuit component has a uniform distribution between 100 and 150, and that these charges are independent of each other for different operations. If the machine is operated five times, what is the probability that the residual charge is between 100 and 110 exactly two times, between 110 and 130 exactly two times, and between 130 and 150 exactly one time?



Can be solved  
using  
multinomial  
distribution

$$X_1 = 2, X_2 = 2, X_3 = 1$$

$$n = 5$$

$$p_1 = 10 \cdot \frac{1}{50} = \frac{1}{5}$$

$$p_2 = 20 \cdot \frac{1}{50} = \frac{2}{5}$$

$$p_3 = 20 \cdot \frac{1}{50} = \frac{2}{5}$$

$$\Rightarrow P(X_1=2, X_2=2, X_3=1) = \frac{5!}{2!2!1!} \cdot \left(\frac{1}{5}\right)^2 \cdot \left(\frac{2}{5}\right)^2 \cdot \frac{2}{5} =$$

$$= \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 2 \cdot 1} \cdot \frac{1}{5} \cdot \frac{1}{5} \cdot \frac{4}{5} \cdot \frac{2}{5} = \frac{48}{625} \approx 0.0768$$